Project Title:
Practical Software Testing: E-Commerce and Student Management Applications.

Project Overview:
This project is designed to give students hands-on experience in manual and automated testing across two types of real-world applications: an e-commerce platform and a student management system. Learners will practice creating test cases, performing functional, integration, and database testing, reporting defects, and applying automation tools where applicable.

Applications for Test:

1. E-Commerce Application: SauceDemo
   Swag Labs

Login Credentials: To access the site, you can use the following
Usernames:

- standard_user
- locked_out_user
- problem_user
- performance_glitch_user

Password: secret_sauce

2. Student Management System: RosarioSIS
   RosarioSIS | Free Student Information System for school management

Objectives:

- Develop proficiency in manual and automated testing.
- Gain experience in functional, integration, regression and UI testing.
- Practice writing clear and detailed test cases.
- Learn to track and report defects effectively using tools like JIRA.
- Apply end-to-end test scenarios for real-world applications.

- Validate data integrity and workflow consistency across multiple modules.
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Tools Required:

- Manual Testing: MS Excel
- Bug Tracking: JIRA.
- Automation: Selenium WebDriver (Python) for web testing.
- API Testing: Postman for API calls (SauceDemo).
- JMETER (SauceDemo).

To Do:

Part 1: E-Commerce Application (SauceDemo)

Modules to Test:

- User Login/Authentication
- Product Listing & Filtering
- Add to Cart & Checkout
- Payment Simulation
- Logout & Session Handling

Testing Activities

1. Functional Testing: Verify login, product search, cart management, checkout.
2. UI/UX Testing: Check layout, responsiveness, navigation and error messages.
3. Regression Testing: Ensure new changes (like product addition) don't break existing flows.
4. Automation Testing: Automate login using Selenium.

Deliverables:

- Test Case Document (MS Excel)
- Bug Reports in JIRA
- Automation Scripts (Selenium)

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Part 2: Student Management System (RosarioSIS)

Modules to Test:

- Student Registration & Profile Management
- Course Enrollment
- Exam Result Entry & Reporting
- Attendance Tracking
- Role-Based Access (Admin, Teacher, Student)

Testing Activities

1. Unit Testing: Validate individual forms (student registration, grade entry).
2. Functional Testing: Verify workflows for enrollment results and attendance.
3. Usability Testing: Evaluate form clarity, navigation and user experience.
4. Integration Testing: Ensure student data flows correctly between enrollment, results and attendance modules.

Deliverables:

- Test Case Document (MS Excel)
- Defect Reports (JIRA

Part 3: Acceptance Test-Driven Development (ATDD)

Feature: Attendance Tracking (RosarioSIS)

Acceptance Criteria:

- Teachers can mark attendance for a registered student.
- Attendance must be shown correctly in student's profile.
- Reports must calculate present/absent totals accurately.

ATDD Test Steps:

1. Login as Teacher.
2. Navigate to Attendance Module.
3. Select Class Grade 10 - Science.
4. Mark Student Bridget Mahlangu as Present.
5. Verify attendance record is saved.
6. Login as Admin → Run attendance report.
7. Confirm Bridget Mahlangu shows as present in today's report.

Template:

Feature	(e.g., Shopping Cart Total Calculation)
User Story	As a, I want , so that .
Acceptance Criteria	
Preconditions	
Test Steps	
Expected Results	
Status	

Part 4: Psychology of Testing-Bias Awareness Exercise

1. Test SauceDemo login with both valid and invalid inputs.
2. Write 5+ negative test cases (for example blank password, special characters, SQL injection attempts).
3. Reflection:
    1. Did you only test happy paths first?
    2. How would a developer vs tester approach differ?
    3. How confirmation bias can affect defect detection.

Deliverable:

- Reflection notes (1 page)
- Negative test case list

Part 5: GitHub CI/CD Integration

Pipeline Setup:

- Push Selenium automation tests to GitHub.
- Create GitHub Actions workflow to:
    1. Run Selenium tests (pytest)
    2. Trigger SonarQube/SonarCloud scan

Deliverables:

- GitHub Repo with CI workflow.
- Screenshot of CI/CD run (all checks passed).

Part 6: Test Levels

1. Define Unit, Integration, System, and Acceptance Testing.
2. Map each test level to one specific activity in SauceDemo and RosarioSIS.
   Example:

- Integration Testing → Verify that items added to cart in SauceDemo are correctly calculated in the checkout total.
- System Testing → Verify full workflow from student registration to exam result generation in RosarioSIS.

Part 7: STLC & SDLC

1. List and explain STLC phases.
2. Map your QA activities to SDLC phases for each project:

- SauceDemo → Login → Cart → Checkout → Payment.
- RosarioSIS → Student Registration → Course Enrollment → Attendance → Result Reporting.

Part 8: API Testing (Postman) – SauceDemo

Test backend endpoints to ensure correct login, product listing, cart management and checkout functionality without relying on the UI.

Key Endpoints:

Feature	Working Endpoint	Method	Notes
Login	https://dummyjson.com/auth/login	**POST**	Test valid & invalid credentials
Product Listing	https://dummyjson.com/products	**GET**	Verify correct product data returned
Add to Cart	https://dummyjson.com/carts/add	**POST**	Ensure items are added with correct quantities
View Cart	https://dummyjson.com/carts/user/1	**GET**	Confirm cart contents

- Create a New Collection called SauceDemo API Tests.
- Add requests for Login, Products, Cart, Checkout.
- Write test scripts in Postman for response code (200, 401), JSON structure and data validation.

Deliverables:

- Screenshots of executed requests (success & negative scenarios)
- Short report summarizing:
  - Endpoints tested
  - Positive & negative scenarios
  - Response validations (status codes, JSON structure, data integrity)

Part 9: Performance / Load Testing (JMeter) – SauceDemo

Simulate multiple users interacting with the application to test performance under load.

Task:

1. **Open JMeter → Test Plan → Thread Group**
   - Number of Threads (Users): 50–100
   - Ramp-Up Period: 10–20 seconds
   - Loop Count: 2
2. **Add HTTP Request Sampler**
   - Server Name: www.saucedemo.com
   - Path: / or /api/auth/login for login test
   - Method: POST/GET depending on API
3. **Add Listeners**
   - View Results Tree
   - Summary Report
   - Aggregate Report
4. **Run Test**
   - Analyze response times, throughput, and errors.
   - Identify bottlenecks (e.g., login delay, slow checkout).

Deliverables:

- Screenshots of JMeter setup (Thread Group, HTTP Samplers, Listeners)
- Performance Reports: Aggregate Report, Summary Report, Graph Results
- Key Findings Document (1 page) summarizing:
  - Average response times
  - Peak concurrent users handled
  - Error rates (if any)
  - Bottlenecks identified

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Part 10: Reflection

1. Reflect on psychology of testing while testing SauceDemo vs RosarioSIS.
2. How did applying STLC improve your testing process across both projects?
3. If you were a QA engineer in a company, how would you apply lessons learned from these two projects?

Submission Guidelines: Project

1. Submit your final work as a PDF document to your LMS Tool.
2. This Activity is an individual Activity
3. Title your document as “Project”
4. Ensure that all questions are answered accordingly. Please do not change/delete or modify the questions and numbers in any way.
5. Include all parts of the activity based on the instructions provided.
6. POWERPOINT PRESENTATIONS ACCEPTED
7. NO CHATGPT ASSISTANCE, Zero Plagiarism Tolerance (Please remember that utilising AI to provide answers is a disservice to you, as these activities are structured to give you hands-on practical experience. Praesignis reserves the right to verify submissions using Plagiarism and AI Detection Tools.

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Project Rubric:

Criteria	Excellence (High-Level Indicators)	Weight (%)
E-Commerce Testing (SauceDemo)	Complete test cases for all modules, defects logged clearly, automation functional	15
Student Management System Testing (RosarioSIS)	Comprehensive test cases, defects reported with severity, integration verified	15
ATDD Scenarios	Acceptance criteria fully mapped, test steps clear	10
Psychology of Testing / Negative Test Cases	Insightful reflection, >5 negative test cases, bias awareness demonstrated	10
GitHub CI/CD Integration	Workflow implemented, Selenium tests and SonarQube scans run successfully	10
Test Levels Mapping	Correct definitions and mapping for both projects	5
STLC & SDLC Mapping	Phases explained and QA activities mapped correctly	5
Reflection / Lessons Learned	Insightful comparison, lessons applied from STLC, SDLC, and psychology	5
Documentation &Presentation	Clear, well-organized, and professional formatting	10
Overall Completeness / Timeliness	All deliverables submitted on time and complete	15

Grading Guide:

90–100% → A
80–89% → B
70–79% → C
60–69% → F

Message to Learners:
Completing this Project gives you hands-on experience in real-world software testing practices. Focus on quality, clarity and reflection, these are the skills that will set you apart as a confident and industry-ready QA professional.
Good luck and enjoy the learning journey!